SAPPhiRE IQ

PROJECT TITLE

Smoking and Health Analysis Dashboard

Domain: Healthcare Analytics

Tool: Microsoft Power BI

Ricky is working as a Health Data Analyst at Sapphire IQ and is responsible for preparing a Smoking and Health Analysis Report for the management team. The management wants to understand patients' smoking habits, health conditions, and key health indicators such as cholesterol and blood pressure.

Ricky needs to develop an interactive Smoking and Health Analysis Dashboard in Power BI using patient-level data. The dashboard will help management monitor key patient KPIs, analyze smoking patterns, compare health conditions across different organs, and identify trends across age groups, gender, and regions.

The report will provide a clear overview of patient health indicators and support data-driven decision-making. It is designed for educational and exploratory analysis and does not establish medical diagnoses or prove that smoking causes specific health conditions.

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PROJECT OBJECTIVES

Objective 1: Analyze Overall Patient Health Using KPIs

1. Calculate the total number of patients.
2. Calculate the total number of current smokers.
3. Calculate the total number of patients with a recorded organ category.
4. Calculate the total number of patients with a recorded health condition.

Objective 2: Create Calculated Columns Using DAX

1. Age Group: Create age categories to group patients into different age ranges, such as 18–29, 30–44, 45–59, and 60+.
2. BP Risk Category: Create categories based on systolic and diastolic blood pressure readings, such as Lower Range, Elevated, and High.

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ANALYSIS & DEMOGRAPHICS

Objective 3: Analyze Smoking Patterns and Health Indicators

1. Analyze average cigarettes consumed per day by smoking status.
2. Analyze organ-wise health condition distribution.
3. Analyze smoking status distribution among patients.
4. Compare average cholesterol levels across blood pressure risk categories.

Objective 4: Analyze Patient Demographics and Health Patterns

1. Analyze the number of current smokers across different age groups.
2. Analyze the distribution of recorded health conditions across different levels of estimated smoking exposure.